

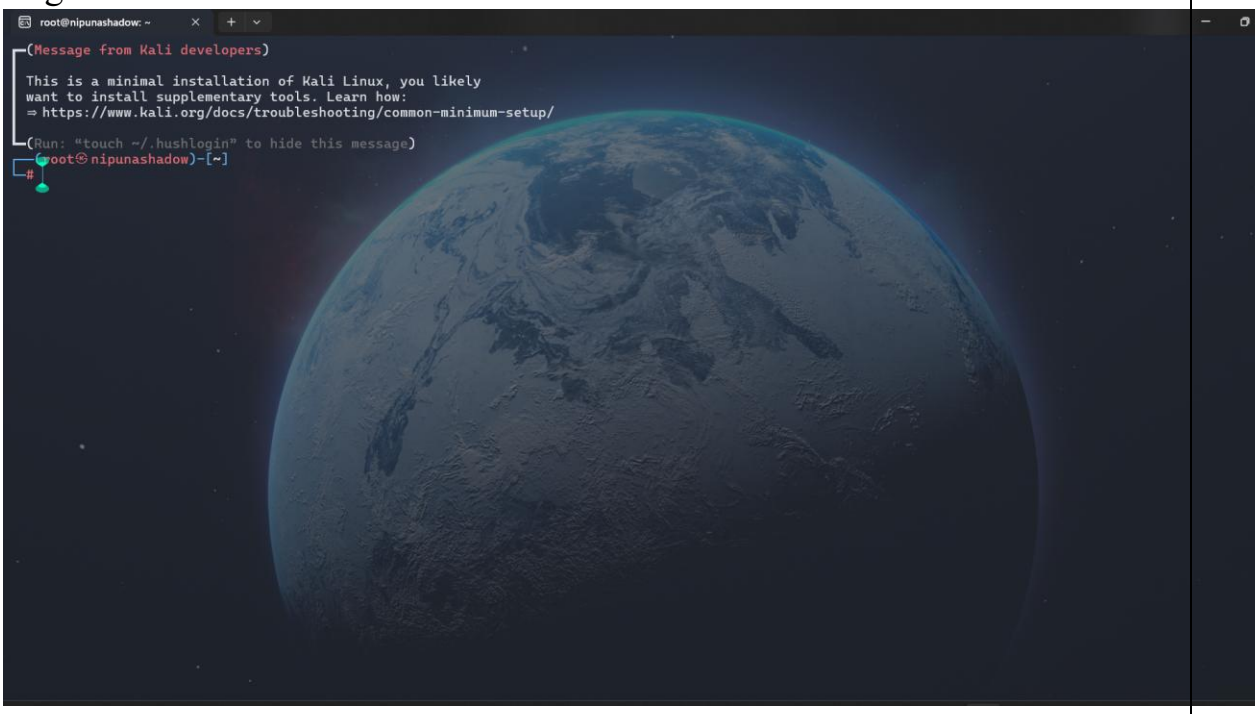
Systems Administration-IIN3462

Lab 01

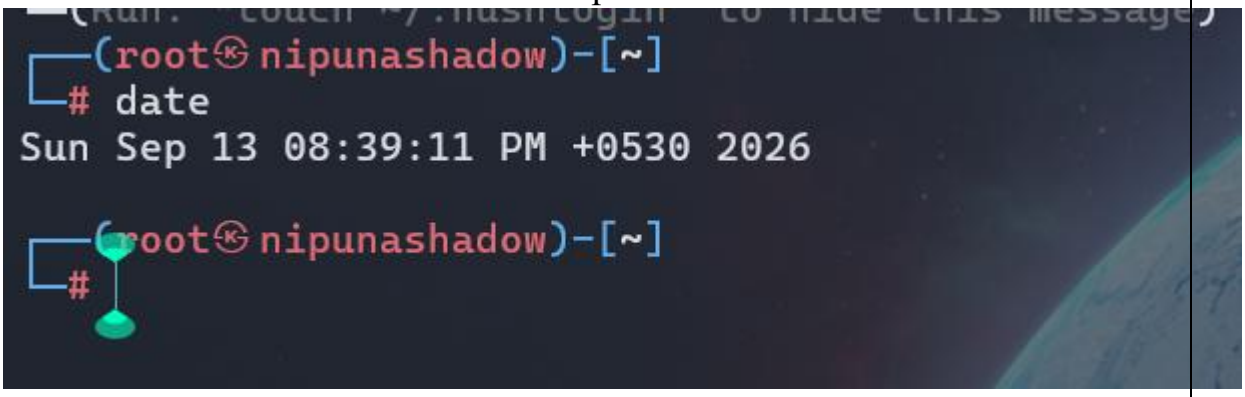
Index No. – ICT/23/810

Name – A.K. Nipuna Darshana

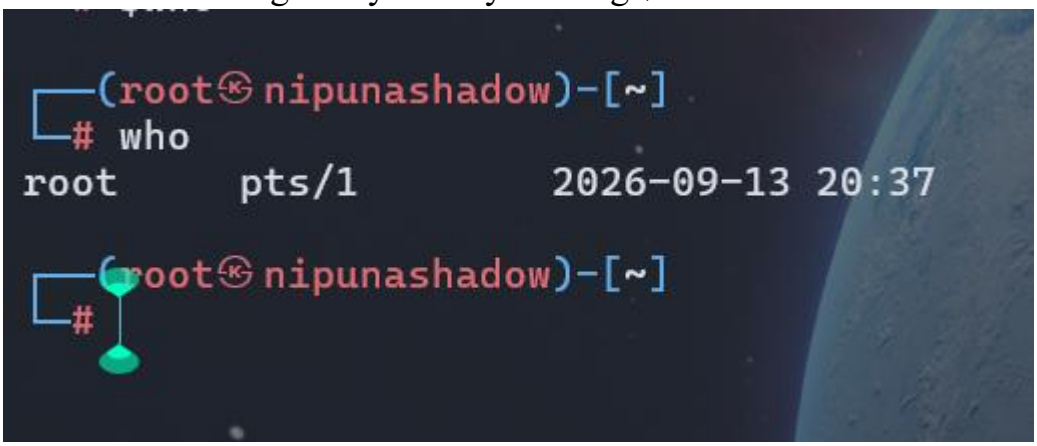
1. Login to Linux server and run terminal.



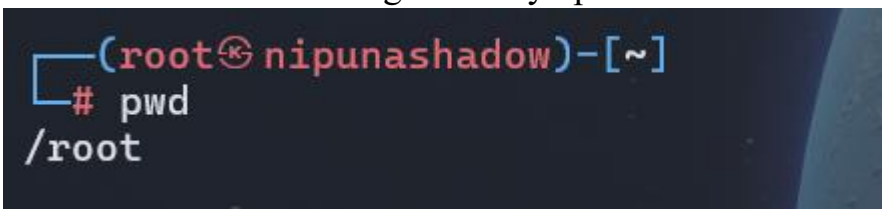
2. Run $date command and record the output



3. Check who's using the system by running $who



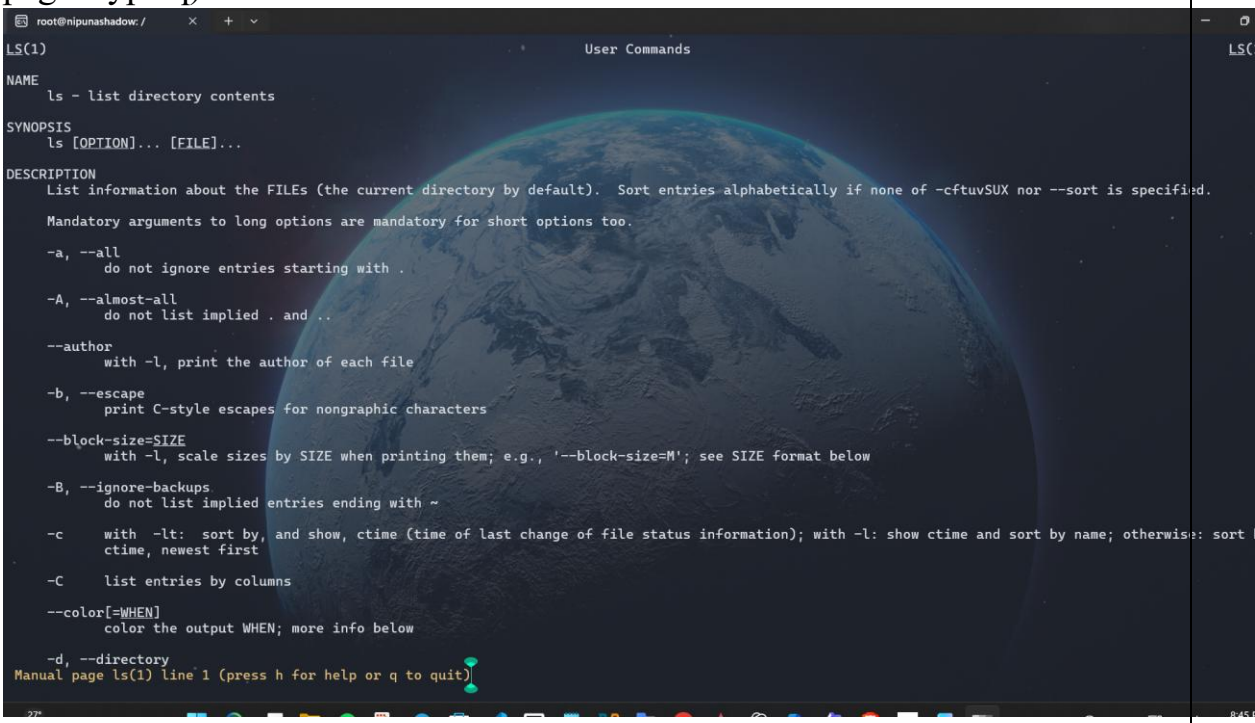
4. Check the current working directory $pwd



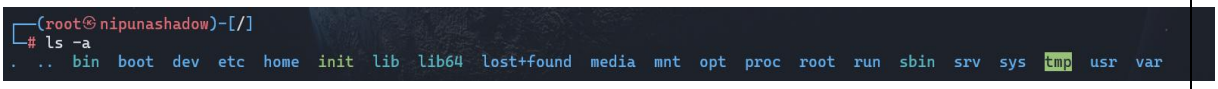
5. List down the content of the folder $ls



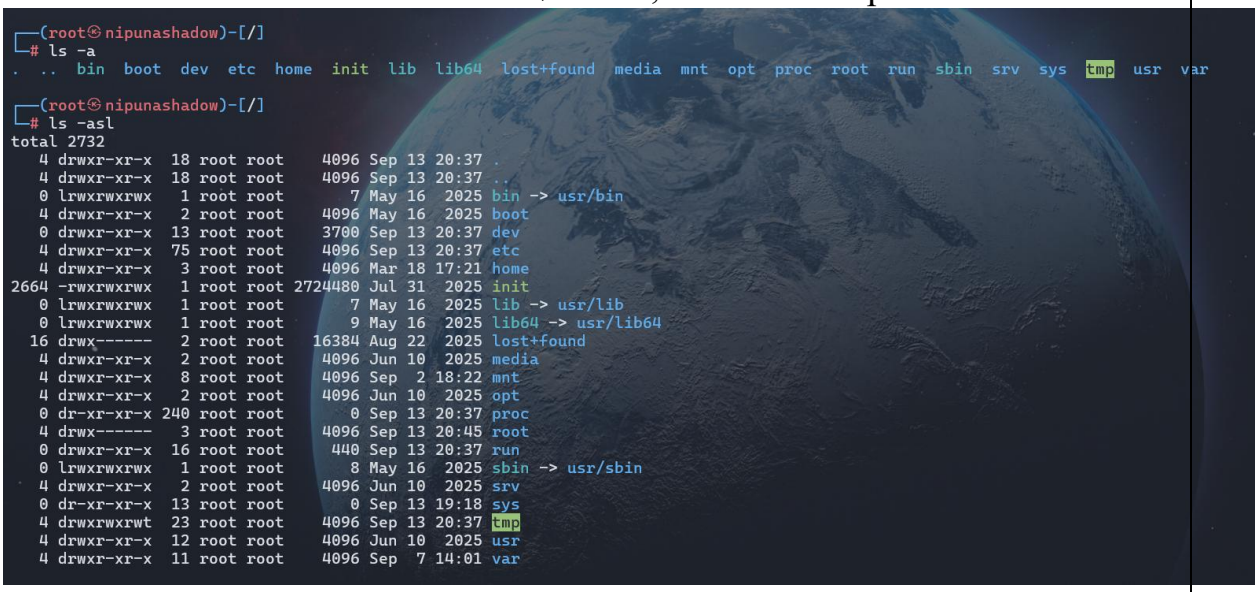
6. View the manual pages for ls command $man ls (For exit form manual pages type q)



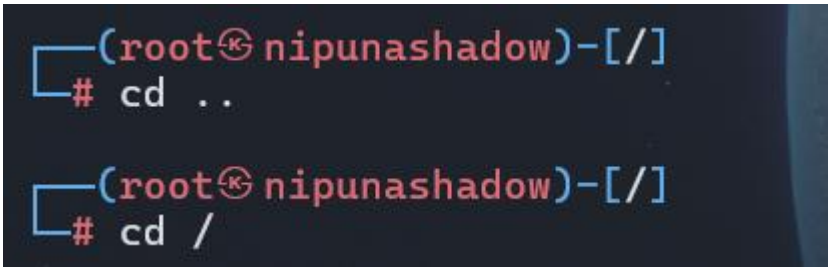
7. What is the option we have to use to show hidden files with $ls command?



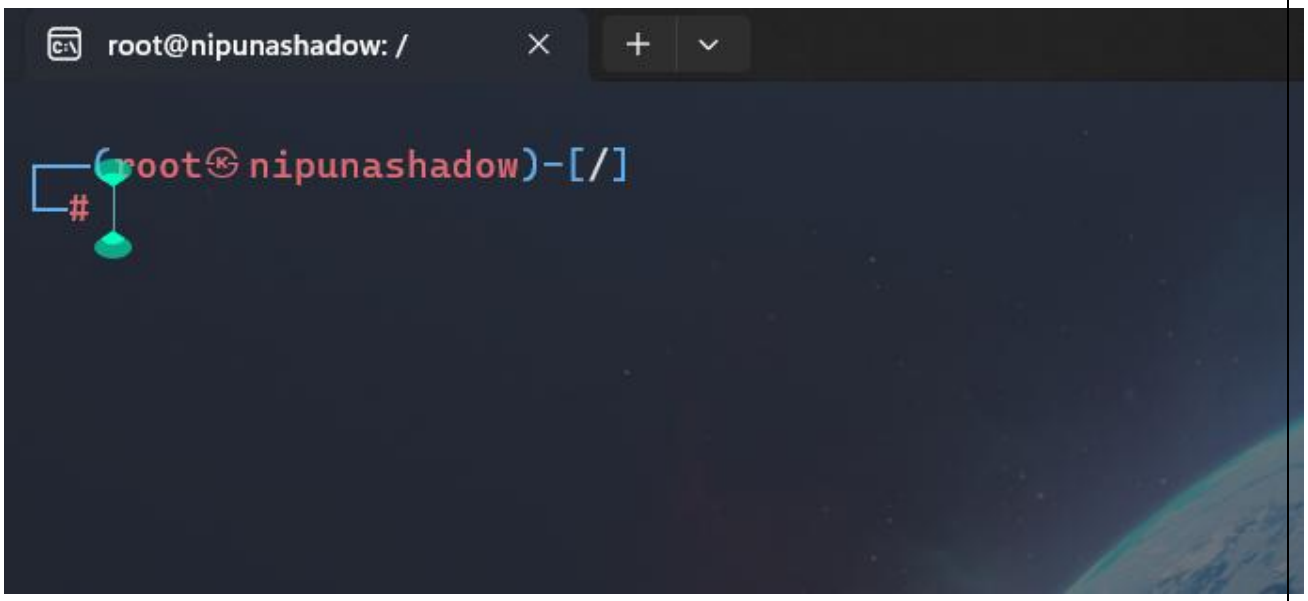
8. Understand the difference between $ls- asl ,describe the options



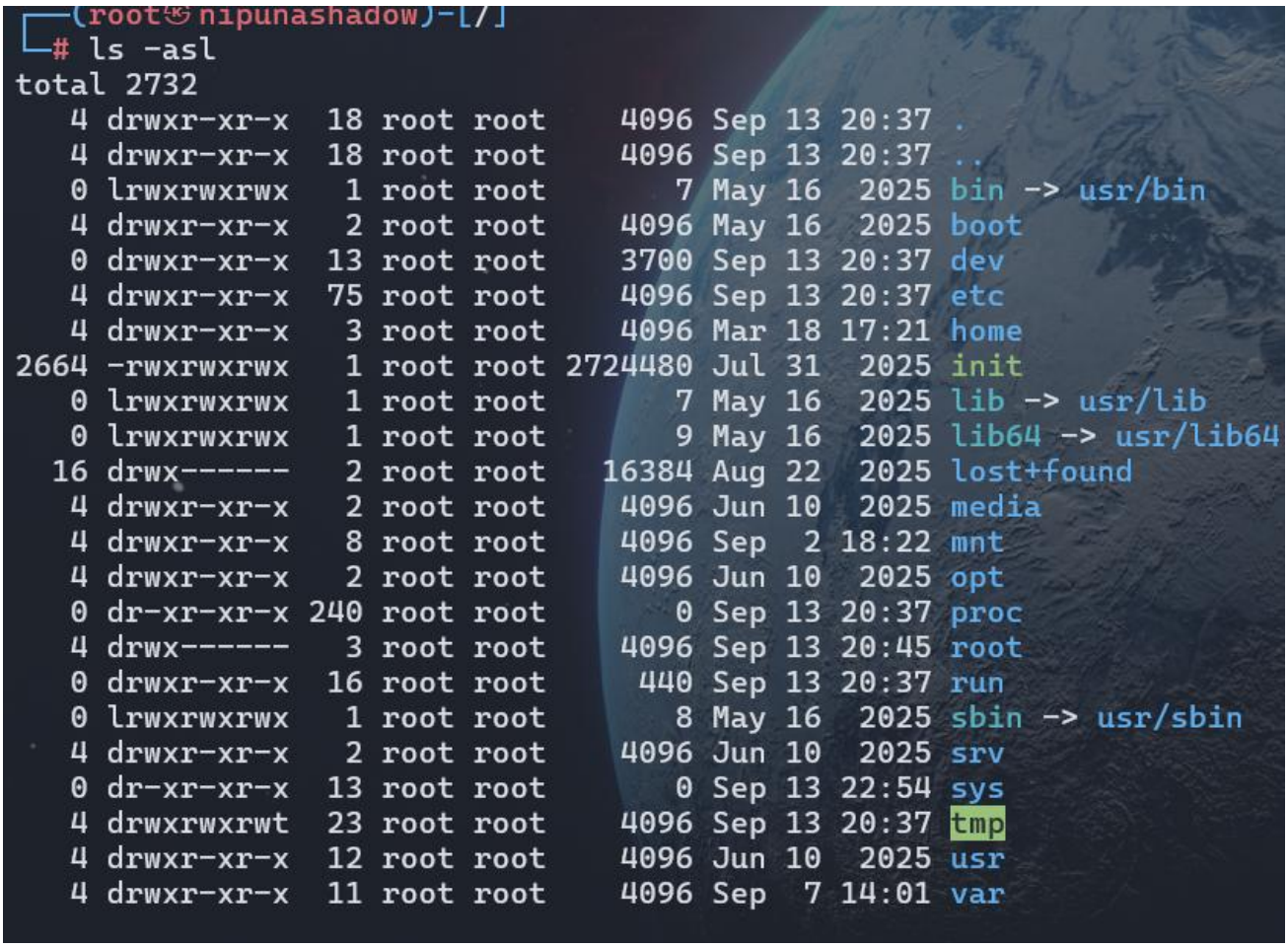
9. Change directory to root $cd /



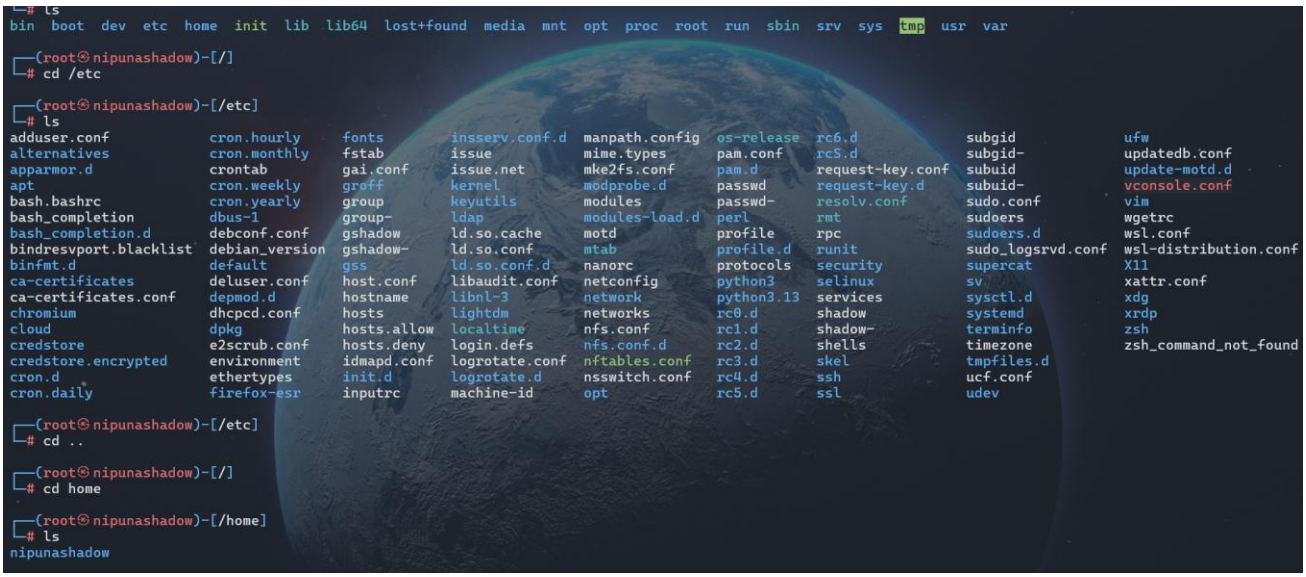
10. Clear Terminal $clear



11. List the content of / partition and record the folder structure of the root partition $ls -asl



12. Go to /etc directory /home /sbin / and observer the content



```
┌──(root㉿nipunashadow)-[/home]
└─# cd ..

┌──(root㉿nipunashadow)-[/]
└─# cd sbin

┌──(root㉿nipunashadow)-[/sbin]
└─# ls
accessdb            delgroup            fsck.cramfs  insmod           mkfs.ext3          parted              rpc.svcgssd  telinit
addgroup            deluser             fsck.ext2    installkernel    mkfs.ext4          partprobe           rtacct       tipc
add-shell           depmod              fsck.ext3    invoke-rc.d      mkfs.minix         plipconfig          rtcwake      tune2fs
adduser             devlink             fsck.ext4    ip               mkhomedir_helper   plocate-build       rtmon        umount.nfs
agetty              dhcpcd              fsck.minix   ipmaddr          mklost+found       poweroff            runlevel     umount.nfs4
arp                 dmidecode           fsfreeze     iptunnel         mkswap             pwck                runuser      unix_chkpwd
arpd                dmsetup             fstab-decode iw               modinfo            pwconv              service      unix_update
badblocks           dmstats             fstrim       iwconfig         modprobe           pwhistory_helper    setcap       update-ca-certificates
biosdecode          dpkg-preconfigure   genl         iwevent          mount.drvfs        pwunconv            sfdisk       update-command-not-found
blkdeactivate       dpkg-reconfigure    getcap       iwgetid          mount.nfs          rarp                shadowconfig updatedb.plocate
blkdiscard          dumpe2fs            getpcaps     iwlist           mount.nfs4         readprofile         showmount    update-passwd
blkid               e2freefrag          getty        iwpriv           mountstats         reboot              shutdown     update-rc.d
blkmapd             e2fsck              groupadd     iwspy            nameif             remove-shell        slattach     update-shells
blkpr               e2image             groupdel     key.dns_resolver netdiscover        request-key         sm-notify    useradd
blockdev            e2label             groupmod     killall5         newusers           resize2fs           sshd         userdel
bridge              e2mmpstatus         grpck        ldconfig         nfsconf            rmmod               start-statd  usermod
capsh               e2scrub             grpconv      logrotate        nfsidmap           rmt                 start-stop-daemon vdpa
cfdisk              e2scrub_all         grpunconv    logsave          nfsiostat          rmt-tar             sudo_logsrvd vigr
chgpasswd           e2undo              halt         losetup          nfsstat            route               sudo_sendlog vipw
chpasswd            e4crypt             hwclock      lsmod            nft                rpcbind             sulogin      visudo
chroot              e4defrag            iconvconfig  mii-tool         nologin            rpcctl              swaplabel    vpddecode
cron                faillock            ifconfig     mke2fs           ownership          rpcdebug            swapoff      wipefs
ctrlaltdel          fdisk               ifdown       mkfs             pam-auth-update    rpc.gssd            swapon       zic
dcb                 filefrag            ifquery      mkfs.bfs         pam_getenv         rpc.idmapd          sysctl       zramctl
debian-update-rc.d  findfs              ifup         mkfs.cramfs      pam_namespace_helper rpcinfo           tarcat
```

13. Go to /dev and observe the content of the dev folder

```
┌──(root㉿nipunashadow)-[/]
└─# cd dev

┌──(root㉿nipunashadow)-[/dev]
└─# ls
autofs          hugepages  log          mptctl      ram12  rtc      stdin   tty18  tty3   tty41  tty53  tty8     vcs1  vcsu
block           hvc0       loop0        mqueue      ram13  rtc0     stdout  tty19  tty30  tty42  tty54  tty9     vcs2  vcsu1
bsg             hvc1       loop1        net         ram14  sda      tty     tty2   tty31  tty43  tty55  ttyS0    vcs3  vcsu2
char            hvc2       loop2        null        ram15  sdb      tty0    tty20  tty32  tty44  tty56  ttyS1    vcs4  vcsu3
console         hvc3       loop3        ppp         ram2   sdc      tty1    tty21  tty33  tty45  tty57  ttyS2    vcs5  vcsu4
core            hvc4       loop4        ptmx        ram3   sdd      tty10   tty22  tty34  tty46  tty58  ttyS3    vcs6  vcsu5
cpu_dma_latency hvc5       loop5        ptp0        ram4   sg0      tty11   tty23  tty35  tty47  tty59  ttyS4    vcsa  vcsu6
disk            hvc6       loop6        ptp_hyperv  ram5   sg1      tty12   tty24  tty36  tty48  tty6   ttyS5    vcsa1 vga_arbi
dxg             hvc7       loop7        pts         ram6   sg2      tty13   tty25  tty37  tty49  tty60  ttyS6    vcsa2 virtio-p
fd              hwrng      loop-control ram0        ram7   sg3      tty14   tty26  tty38  tty5   tty61  ttyS7    vcsa3 vport0p0
full            initctl    mapper       ram1        ram8   shm      tty15   tty27  tty39  tty50  tty62  urandom  vcsa4 vport0p1
fuse            kmsg       mcelog       ram10       ram9   snapshot tty16   tty28  tty4   tty51  tty63  userfaultfd vcsa5 vport0p2
hpet            kvm        mem          ram11       random stderr   tty17   tty29  tty40  tty52  tty7   vcs      vcsa6 vsock
```

14. Describe the output

```
┌──(root㉿nipunashadow)-[~]
└─# cd /
ls -asl
total 2732
   4 drwxr-xr-x  18 root root    4096 Sep 13 20:37 .
   4 drwxr-xr-x  18 root root    4096 Sep 13 20:37 ..
   0 lrwxrwxrwx   1 root root       7 May 16  2025 bin -> usr/bin
   4 drwxr-xr-x   2 root root    4096 May 16  2025 boot
   0 drwxr-xr-x  13 root root    3700 Sep 13 20:37 dev
   4 drwxr-xr-x  75 root root    4096 Sep 13 23:16 etc
   4 drwxr-xr-x   4 root root    4096 Sep 13 23:10 home
2664 -rwxrwxrwx   1 root root 2724480 Jul 31  2025 init
   0 lrwxrwxrwx   1 root root       7 May 16  2025 lib -> usr/lib
   0 lrwxrwxrwx   1 root root       9 May 16  2025 lib64 -> usr/lib64
  16 drwx------   2 root root   16384 Aug 22  2025 lost+found
   4 drwxr-xr-x   2 root root    4096 Jun 10  2025 media
   4 drwxr-xr-x   8 root root    4096 Sep  2 18:22 mnt
   4 drwxr-xr-x   2 root root    4096 Jun 10  2025 opt
   0 dr-xr-xr-x 240 root root       0 Sep 13 20:37 proc
   4 drwx------   3 root root    4096 Sep 13 23:16 root
   0 drwxr-xr-x  16 root root     440 Sep 13 20:37 run
   0 lrwxrwxrwx   1 root root       8 May 16  2025 sbin -> usr/sbin
   4 drwxr-xr-x   2 root root    4096 Jun 10  2025 srv
   0 dr-xr-xr-x  13 root root       0 Sep 13 22:54 sys
   4 drwxrwxrwt  23 root root    4096 Sep 13 20:37 tmp
   4 drwxr-xr-x  12 root root    4096 Jun 10  2025 usr
   4 drwxr-xr-x  11 root root    4096 Sep  7 14:01 var
```

Part	Value	Meaning
d	file type	**d**irectory
r-xr-xr-x	permissions	Owner: read+execute, Group: read+execute, Others: read+execute
root	owner	Owned by user root
root	group	Belongs to group root
4096	size	Size in bytes (4 KB)
Sep 13 20:37	date	Last modified date/time
bin	name	The folder name

15.Create folder in home folder by specifying folder name as Student id number.(Make sure use this folder for store all the files created in

proceeding exercises) $mkdir



16. Learn vi editor by doing following activities. • Create text file name myinfor.txt using following command $vi myinfor.txt

